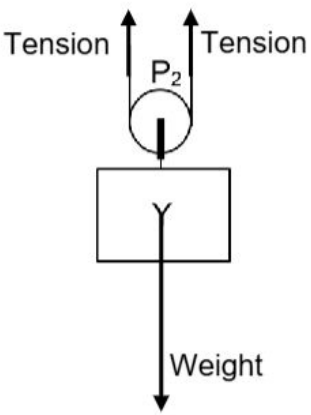
2	**(a)**	Newton's first law of motion states that a body at rest will stay at rest, and a body in motion will continue to move at constant velocity, unless acted on by a resultant external force.		A1
	(b)	**(i)**	work done by $f = 5.0 \times 1.5 \cos 180° = -7.5$ J	A1
		(ii)	By conservation of energy, total kinetic energy = $m_x g\Delta h_x + m_y g\Delta h_y$ + work done by friction $= (2.0)(9.81)(1.5 \cos 60°) + (4.0)(9.81)(0.75) - 7.5$ $= 36.6$ J ≈ 37 J (allow ecf for work done by friction) (allow use of kinematics equations to find velocities and total kinetic energy)	C1 A1
		(iii)	 Tensions and Weight drawn in the correct directions and positions and both labelled. (Do not accept weight labelled as *W*.) (Ignore magnitude of the forces.)	A1
		(iii)	$T + m_x g \cos 60° - f = m_x a_x$ $T + (2.0)g \cos 60° - 5.0 = (2.0)(2a_y)$ --- eqn 1 $m_y g - T - T = m_y a_y$ $4.0g - 2T = (4.0)a_y$ --- eqn 2 Solving, $T = 11.477$ N ≈ 11.5 N or 11 N	M1 M1 A1